Physics 4730, Fall 2026 Final Project

Due Sunday December 13th

For this semester's final project, we will analyze public astrometric data from the European Space Agency's *Gaia* observatory. From this data, we will seek to identify stars located in the Hyades open star cluster[1]. We will make quantitative estimates (with uncertainties) of distance and tangential speed of this cluster, compare relevant distributions with those of background stars, and examine the Hyades cluster in the context of the Hertzprung-Russell diagram[2].

The Data

The file `hyades_gaia_dr3.csv` was obtained by the instructor from a *Gaia* data repository. Please see the appendix for details of the instructor's data query.

The Jupyter Notebook `hyades.ipynb` (assumed to be in the same directory) demonstrates how this file may be read and the data adapted to make use of numpy tools. Becoming familiar with this notebook should be the starting point of your investigations.

The following table lists the data in the .csv file, with a brief description, units and common symbology.

data name	description	units	symbol
`source_id`	*Gaia* identifier	—	—
`ra`	right ascension (RA)	degrees	α
`dec`	declination (DEC)	degrees	δ
`parallax`	parallax	mas[3]	ϖ
`parallax_error`	std uncertainty	mas	
`pmra`	proper motion in RA $\mu_{\alpha*} = \mu_\alpha \cos\delta$	mas/yr	$\mu_{\alpha*}$
`pmra_error`	std uncertainty	mas/yr	—
`pmdec`	proper motion in DEC	mas/yr	μ_δ
`pmdec_error`	std uncertainty	mas/yr	—
`parallax_pmra_corr`	correlation coefficient	—	—
`parallax_pmdec_corr`	correlation coefficient	—	—
`pmra_pmdec_corr`	correlation coefficient	—	—
`phot_g_mean_mag`	mean G band brightness	mag	G
`bp_rp`	*Gaia* color:	mag	$G_{\rm BP} - G_{\rm RP}$
`ruwe`	Renormalized Unit Weight Error, an astrometric fit quality indicator	—	—
`visibility_periods_used`	number of *Gaia* observing periods used in the astrometric solution	—	—

[1] https://en.wikipedia.org/wiki/Hyades_(star_cluster)

[2] https://www.britannica.com/science/Hertzsprung-Russell-diagram

[3] milli-arcseconds

Required Analysis

1. Inspect the data and define samples

Make a scatter plot of proper-motion (μ_δ versus $\mu_{\alpha*}$) and a EPDF histogram of the parallax ϖ. Be sure to explain the meanings of μ_δ, $\mu_{\alpha*}$ and ϖ.

Select your Hyades *candidate* sample by making cuts on proper motion and parallax. The remaining sources should be kept as a *control* sample. You will want to look through the literature to get an idea where to make these cuts: Be sure to cite any references used here and explain your choices. Be sure to state candidate and control sample sizes in prose.

Do try to isolate the candidate sources from the background, but don't try to tune your cuts to get the "right" answers to quantities we will want to find below. The instructor's query criteria were not designed to produce an unbiased membership determination. Can you identify any reasons why?

Now, remake your scatter plot and parallax histogram, clearly identifying the candidate and control sample sources.

2. Propagate measurement errors

For each candidate star calculate it's distance from Earth in in parsecs

$$d = \frac{1}{\varpi}$$

along with its associated uncertainty.

For each candidate star calculate the *tangential speed*

$$v_t = 4.74047 \, \frac{\sqrt{\mu_{\alpha*}^2 + \mu_\delta^2}}{\varpi},$$

where ϖ is in mas and proper motions are in mas/yr; v_t is in km/s. Construct the covariance matrix for $(\varpi, \mu_{\alpha*}, \mu_\delta)$ from *Gaia*'s standard errors and correlation coefficients. Use

$$\sigma_{v_t}^2 = \nabla v_t^{\mathsf{T}} C \nabla v_t.$$

to compute the final uncertainty in tangential velocity.

In a table, display the mean, standard deviation, minimum and maximum values of the distance along with their errors. Also display the mean, standard deviation, minimum and maximum values of the tangential velocity along with their errors.

3. Fit a statistical model

Model candidate parallaxes with a normal distribution $\mathcal{N}$ having mean ϖ_0 and width given by $(\sigma_{\varpi,i}^2 + s_{\rm int}^2)^{1/2}$.

$$\varpi_i \sim \mathcal{N}\left(\varpi_0, \left(\sigma_{\varpi,i}^2 + s_{\rm int}^2\right)^{1/2}\right),$$

where ϖ_0 is the cluster mean parallax, $\sigma_{\varpi,i}$ is the reported uncertainty, and $s_{\rm int} \geq 0$ represents intrinsic scatter. Perform a maximum-likelihood calculation to determine the best value of the intrinsic scatter along with its 68% c.l.uncertainty.

Compute the χ^2 statistic using the fitted intrinsic scatter. Report χ^2, the degrees of freedom, and the associated p-value. Display the standardized-residual Q–Q plot and explain what a very small goodness-of-fit p-value would mean.

4. Compare distributions

We'd like to know whether the sources we've identified as coming from the Hyades cluster are "typical" in terms of quantities associated with measurement quality. Use a two-sample Kolmogorov-Smirnov (KS) test to compare candidate and control datasets in the random variables

- Renormalized Unit Weight Error
- number of observing periods
- standard uncertainty in parallax

For each of these RVs, include an ECDF overlay along with reporting the KS-statistic and p-value for the comparison. Comment on what you find.

5. Hertzprung-Russell Diagram

Finally, display both the candidate sources and control sources on the same color-magnitude (Hertzprung-Russell) diagram. Plot the *Gaia* color $G_{\rm BP} - G_{\rm RP}$ on the horizontal axis against the absolute G-band magnitude

$$M_G = G + 5\log_{10}(\varpi_{mas}) - 10$$

on the vertical axis. Ideally, the stars that are actually part of the Hyades cluster will lie in a much tighter sequence than that of the control group. Try fitting the candidates to a simple function – *e.g.* a cubic or quartic polynomial – to confirm that this is the case.

Notebook and writing requirements

Submit one clearly organized Jupyter Notebook named **final-project.ipynb**. It must run in order without manual intervention, read the data file, mix executable code with explanatory Markdown cells, label all axes and units, define symbols before using them, and report numerical results with sensible precision. Do not merely paste output: explain the statistical reasoning and interpret each result.

Your conclusion must recap the answers to the scientific questions quantitatively, summarizing the statistical evidence. You should also and identify and discuss any limitation(s) to this analysis that you can think of. Feel free to discuss this with the instructor.

Pre-Submission checklist

Item	Complete
Notebook runs from top to bottom	☐
Gaia DR3 and any other source(s) is/are cited	☐
Candidate and control selections are stated and justified	☐
Covariance-aware propagation for v_t is shown	☐
Likelihood model and parameter estimates are explained	☐
Goodness-of-fit statistic and confidence interval are interpreted	☐
KS statistic, p-values, and are discussed	☐
HR diagram demonstrates candidates are in tighter sequence than control	☐
Conclusion answers the scientific question quantitatively	☐

Submission

Submit your final project by 11:59 PM on Sunday, December 13:

submit p4730 final-project final-project.ipynb

Appendix: Details of Obtaining the Data

The data file `hyades_gaia_dr3.csv` to be used in this analysis was obtained by the instructor from a *Gaia* data repository:

https://mast.stsci.edu/vo-tap/api/v0.1/gaiadr3/

A synchronous ADQL (Astronomical Data Query Language) search was conducted with the query

```
SELECT TOP 5000
    source_id, ra, dec,
    parallax, parallax_error,
    pmra, pmra_error,
    pmdec, pmdec_error,
    parallax_pmra_corr,
    parallax_pmdec_corr,
    pmra_pmdec_corr,
    phot_g_mean_mag, bp_rp, ruwe,
    visibility_periods_used
FROM dbo.gaia_source
WHERE 1 = CONTAINS(
    POINT('ICRS', ra, dec),
    CIRCLE('ICRS', 66.5, 16.0, 1.0)
)
AND parallax_over_error > 10
AND visibility_periods_used >= 8
AND ruwe < 1.4
AND phot_g_mean_mag < 17
```

A total of 2,335 quality-screened sources were obtained for the analysis.